

Supplementary Information

Four-Corner Spectral Tuning for Quadratic ADMM with Application to Diffeomorphic Image Registration

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1 Purpose and notation

This supplement develops the optional noncommuting rate certificates, their proofs, scaling diagnostics, and implementation details. These constructions are not used to choose the deployed four-corner parameters. For symmetric positive semidefinite $H, G \in \mathbb{R}^{dn \times dn}$ and $\rho > 0$, define

$$C_A(\rho) = (A - \rho \mathbf{I})(A + \rho \mathbf{I})^{-1},$$

for any symmetric positive semidefinite A , and abbreviate

$$C_H = (H - \rho \mathbf{I})(H + \rho \mathbf{I})^{-1}, \quad C_G = (G - \rho \mathbf{I})(G + \rho \mathbf{I})^{-1}.$$

The reduced oADMM error operator is

$$E_{\rho, \alpha} = \left(1 - \frac{\alpha}{2}\right) \mathbf{I} + \frac{\alpha}{2} C_H C_G, \quad 0 < \alpha \leq 2. \quad (1)$$

Let nonoverlapping patches $\{\Omega_r\}$ partition the full grid, and define the full-size direct-sum surrogate

$$\tilde{H} = \bigoplus_r (I_{|\Omega_r|} \otimes H_r), \quad \tilde{G} = \bigoplus_r (G_r \otimes I_d),$$

where each H_r and G_r is symmetric positive semidefinite. Let

$$\delta_H = \|H - \tilde{H}\|_2, \quad \delta_G = \|G - \tilde{G}\|_2, \quad \delta = \delta_H + \delta_G.$$

The two surrogate operators commute blockwise. If the symbol endpoints used in the modal maximization are attained, its exact frozen-patch envelope satisfies $U_{\text{patch}} = \|\tilde{E}_{\rho, \alpha}\|_2$. The notation “exact” always refers to this stated frozen separable model; the full variable-coefficient registration operator is generally noncommuting.

2 Additive perturbation certificate

2.1 Cayley Lipschitz estimate

Lemma 2.1 (Cayley Lipschitz bound). *For symmetric positive semidefinite A, B and $\rho > 0$,*

$$\|C_A(\rho) - C_B(\rho)\|_2 \leq \frac{2}{\rho} \|A - B\|_2. \quad (2)$$

Proof. Writing $C_A = \mathbf{I} - 2\rho(A + \rho\mathbf{I})^{-1}$ and using the resolvent identity gives

$$\begin{aligned} C_A - C_B &= 2\rho [(B + \rho\mathbf{I})^{-1} - (A + \rho\mathbf{I})^{-1}] \\ &= 2\rho(A + \rho\mathbf{I})^{-1}(A - B)(B + \rho\mathbf{I})^{-1}. \end{aligned}$$

Both resolvent norms are at most ρ^{-1} , proving (2). $\square$

Theorem 2.2 (Additive frozen-patch certificate). *For the direct-sum surrogate defined above and the full variable-coefficient iteration of the same dimension,*

$$\varrho(E_{\rho,\alpha}) \leq U_{\text{patch}}(\rho, \alpha) + \frac{\alpha\delta}{\rho}. \quad (3)$$

No commutativity assumption on H and G is required.

Proof. Since every Cayley transform of a positive semidefinite matrix has norm at most one,

$$\begin{aligned} \|C_H C_G - C_{\tilde{H}} C_{\tilde{G}}\|_2 &\leq \|(C_H - C_{\tilde{H}})C_G\|_2 \\ &\quad + \|C_{\tilde{H}}(C_G - C_{\tilde{G}})\|_2 \\ &\leq \frac{2}{\rho}(\delta_H + \delta_G). \end{aligned}$$

Apply (1), the triangle inequality, and normality of the separable direct-sum operator, for which its operator norm equals U_{patch} . Any cross-patch couplings removed from the regularizer are included in δ_G . $\square$

Corollary 2.3 (Certified linear rate). *If the right-hand side of (3) is below one, then the full linearized oADMM iteration converges linearly with asymptotic factor no larger than that value.*

2.2 Certificate-aware legacy objective

For standard ADMM, all scalar frozen-modal factors are nonnegative. A Let $\mathcal{C}$ denote the finite set of attained frozen-model curvature-symbol corners. A certificate-aware diagnostic objective is therefore

$$\mathcal{U}_1(\rho) = \max_{c \in \mathcal{C}} \frac{\rho^2 + h_c g_c}{(\rho + h_c)(\rho + g_c)} + \frac{\delta}{\rho}. \quad (4)$$

For two corner branches $c_1 = (h_1, g_1)$ and $c_2 = (h_2, g_2)$, equality after denominator clearing is

$$\rho(A_{12}\rho^2 + B_{12}) = 0,$$

where

$$A_{12} = (h_2 + g_2) - (h_1 + g_1), \quad B_{12} = g_1 g_2 (h_1 - h_2) + h_1 h_2 (g_1 - g_2).$$

On an interval where $c = (h, g)$ is active, stationarity of (4) gives

$$(h + g)\rho^2(\rho^2 - hg) - \delta(\rho + h)^2(\rho + g)^2 = 0. \quad (5)$$

Thus the legacy standard-ADMM bound can also be minimized over a finite set of endpoints, branch intersections, and active quartic roots. This diagnostic is not the deployed four-corner selector: the additive term often moves its minimizer toward an unhelpful boundary.

For an admissible interval $[\alpha_-, \alpha_+] \subset (0, 2]$, let a and b be the active minimum and maximum frozen-modal θ values at the selected penalty. The certificate-aware relaxation objective is

$$F(\alpha) = \max\{|1 - \alpha + \alpha a|, |1 - \alpha + \alpha b|\} + \frac{\alpha \delta}{\rho}.$$

It is convex and piecewise linear. A minimizer belongs to

$$\left\{ \alpha_-, \alpha_+, \frac{1}{1-a}, \frac{1}{1-b}, \frac{2}{2-a-b} \right\} \cap [\alpha_-, \alpha_+].$$

3 Exact-regularizer block structure

The additive certificate replaces both operators. Sharper bounds retain the regularizer exactly. Let

$$G = G_0 \otimes I_d, \quad K_\rho = (G_0 - \rho I_n)(G_0 + \rho I_n)^{-1},$$

so $C_G = K_\rho \otimes I_d$. Write

$$C_H = \text{diag}(C_1, \dots, C_n), \quad C_i = (H_i - \rho I_d)(H_i + \rho I_d)^{-1}.$$

Then the (i, j) block of $M := C_H C_G$ is

$$M_{ij} = (K_\rho)_{ij} C_i. \tag{6}$$

Although K_ρ and every C_i are symmetric, M is generally nonnormal because the C_i vary.

4 Block-Gershgorin disks

Theorem 4.1 (Block-Gershgorin enclosure). *Every eigenvalue of $M = C_H C_G$ lies in a disk*

$$|z - (K_\rho)_{ii} c_{i,r}| \leq \|C_i\|_2 \sum_{j \neq i} |(K_\rho)_{ij}| \tag{7}$$

for some pixel i and eigenvalue $c_{i,r}$ of C_i . Consequently

$$U_{\text{BG}}(\rho, \alpha) = \max_{i,r} \left[\left| 1 - \frac{\alpha}{2} + \frac{\alpha}{2} (K_\rho)_{ii} c_{i,r} \right| + \frac{\alpha}{2} \|C_i\|_2 \sum_{j \neq i} |(K_\rho)_{ij}| \right] \tag{8}$$

is a rigorous upper bound on $\varrho(E_{\rho, \alpha})$.

Proof. Let $Mx = zx$ and choose a block x_i of maximum Euclidean norm. The i th block equation is

$$(zI_d - (K_\rho)_{ii} C_i) x_i = \sum_{j \neq i} (K_\rho)_{ij} C_i x_j.$$

The diagonal block is normal, so its smallest singular value is the distance from z to its spectrum. Taking norms, using $\|x_j\| \leq \|x_i\|$, and cancelling $\|x_i\|$ gives (7). The affine map from M to $E_{\rho, \alpha}$ maps each disk to the bound (8). $\square$

5 Signed rectangle and commutator information

Decompose

$$S = \frac{M + M^\top}{2}, \quad Q = \frac{M - M^\top}{2}.$$

For $i \neq j$,

$$S_{ij} = (K_\rho)_{ij} \frac{C_i + C_j}{2}, \quad Q_{ij} = (K_\rho)_{ij} \frac{C_i - C_j}{2}.$$

Define zero-diagonal symmetric comparison matrices by

$$B_{ij}^S = |(K_\rho)_{ij}| \|(C_i + C_j)/2\|_2, \quad B_{ij}^Q = |(K_\rho)_{ij}| \|(C_i - C_j)/2\|_2, \quad (9)$$

and let d_i^- and d_i^+ be the extreme eigenvalues of the diagonal block $(K_\rho)_{ii}C_i$. The block-row quantities are

$$m_\rho^{\text{row}} = \min_i \left[d_i^- - \sum_{j \neq i} B_{ij}^S \right], \quad (10)$$

$$M_\rho^{\text{row}} = \max_i \left[d_i^+ + \sum_{j \neq i} B_{ij}^S \right], \quad (11)$$

$$\eta_\rho^{\text{row}} = \max_i \sum_{j \neq i} B_{ij}^Q. \quad (12)$$

Theorem 5.1 (Commutator-aware signed rectangle). *With $W(M) = \{x^* M x : \|x\|_2 = 1\}$ denoting the numerical range,*

$$W(M) \subset [m_\rho^{\text{row}}, M_\rho^{\text{row}}] + i[-\eta_\rho^{\text{row}}, \eta_\rho^{\text{row}}].$$

Therefore

$$\varrho(E_{\rho, \alpha}) \leq U_{\text{rect}} := \max_{x \in \{m_\rho^{\text{row}}, M_\rho^{\text{row}}\}} \sqrt{\left(1 - \frac{\alpha}{2} + \frac{\alpha x}{2}\right)^2 + \left(\frac{\alpha \eta_\rho^{\text{row}}}{2}\right)^2}. \quad (13)$$

Proof. For any normalized complex vector x , set $z = x^* M x$. Then

$$\text{Re } z = x^* S x, \quad \text{i Im } z = x^* Q x.$$

Hermitian block-Gershgorin bounds applied to S put the first quantity in the stated real interval. The symmetric matrix of block norms bounds $\|Q\|_2$ by its maximum row sum, giving $|\text{Im } z| \leq \eta_\rho^{\text{row}}$. Applying the affine map in (1) and maximizing the Euclidean modulus over a rectangle gives (13); convexity puts the real-coordinate maximum at an endpoint. $\square$

For fixed ρ , the squared expressions under the maximum in (13) are quadratics in α . Their minimum is attained at an admissible interval endpoint, a stationary point of either quadratic, or their intersection

$$\alpha = \frac{4}{2 - m_\rho^{\text{row}} - M_\rho^{\text{row}}}$$

when admissible. This finite conditional selection is useful for analyzing a certificate but is not used in one-shot registration tuning.

6 Perron comparison rectangle

The row maxima can be replaced by global scalar eigenproblems.

Theorem 6.1 (Perron comparison). *The signed rectangle theorem remains valid with*

$$\begin{aligned} m_\rho &= \lambda_{\min}(\text{diag}(d^-) - B^S), \\ M_\rho &= \lambda_{\max}(\text{diag}(d^+) + B^S), \\ \eta_\rho &= \varrho(B^Q). \end{aligned} \tag{14}$$

These bounds are no weaker than their block-row counterparts. Denote the resulting value of (13) by U_{cmp} .

Proof. Partition a complex block vector x and set $y_i = \|x_i\|_2$. For the symmetric part,

$$x^* S x \leq \sum_i d_i^+ y_i^2 + 2 \sum_{i < j} B_{ij}^S y_i y_j = y^\top (\text{diag}(d^+) + B^S) y.$$

The analogous lower estimate uses $\text{diag}(d^-) - B^S$. Rayleigh–Ritz yields the first two bounds. Blockwise comparison for the skew part gives

$$\|Qx\|_2 \leq \|B^Q y\|_2,$$

and hence $\|Q\|_2 \leq \|B^Q\|_2 = \varrho(B^Q)$ because B^Q is symmetric and entrywise nonnegative. Finally, extremal eigenvalues and spectral radii of the comparison matrices are bounded by their maximum absolute row sums, proving dominance over the row construction. $\square$

The comparison method requires three symmetric scalar $n \times n$ extremal eigenproblems and does not form the full $dn \times dn$ oADMM matrix.

7 Sparse truncated-kernel certificate

Under periodic boundaries, K_ρ is convolution by a scalar discrete kernel $k_\rho(s)$. Let $\mathcal{R} = -\mathcal{R}$ be a retained finite symmetric set of offsets containing zero, and define $K_{\rho, \mathcal{R}}$ as convolution by $k_\rho(s) \mathbf{1}_{\{s \in \mathcal{R}\}}$. Define

$$\tau_{\mathcal{R}} = \sum_{s \notin \mathcal{R}} |k_\rho(s)|.$$

Let $M_{\mathcal{R}} = C_H(K_{\rho, \mathcal{R}} \otimes I_d)$. Construct the Perron comparison matrices using retained offsets only and let $U_{\mathcal{R}}$ be the resulting affine numerical-range rectangle bound from Theorem 6.1.

Corollary 7.1 (Truncated convolution certificate).

$$\varrho(E_{\rho, \alpha}) \leq U_{\mathcal{R}} + \frac{\alpha}{2} \max_i \|C_i\|_2 \tau_{\mathcal{R}}. \tag{15}$$

Proof. Young’s convolution inequality gives

$$\|K_\rho - K_{\rho, \mathcal{R}}\|_2 \leq \tau_{\mathcal{R}}.$$

Write $M = M_{\mathcal{R}} + D$. Left multiplication by C_H gives

$$\|D\|_2 \leq \max_i \|C_i\|_2 \tau_{\mathcal{R}}.$$

For a normalized eigenvector x of M with eigenvalue z , $z = x^* M_{\mathcal{R}} x + x^* D x$. The retained numerical-range rectangle bounds

$$\left| 1 - \frac{\alpha}{2} + \frac{\alpha}{2} x^* M_{\mathcal{R}} x \right| \leq U_{\mathcal{R}},$$

while $|x^* D x| \leq \|D\|_2$. The triangle inequality yields (15). $\square$

For fixed truncation radius, the comparison matrices are sparse. The kernel can be evaluated by FFT in $O(n \log n)$ time and extremal comparison eigenvalues can be estimated matrix-free. The bound is rigorous at every radius, including zero; usefulness depends on tightness.

8 Commutator and orientation diagnostics

The resolvent identity also yields a direct commutator estimate.

Proposition 8.1 (Cayley commutator bound).

$$\|[C_H, C_G]\|_2 \leq \frac{4\rho^2 \|[H, G]\|_2}{(\lambda_{\min}(H) + \rho)^2 (\lambda_{\min}(G) + \rho)^2}. \quad (16)$$

Proof. Use $C_H = I - 2\rho(H + \rho I)^{-1}$ and the identity

$$[(H + \rho I)^{-1}, (G + \rho I)^{-1}] = (H + \rho I)^{-1} (G + \rho I)^{-1} [H, G] (G + \rho I)^{-1} (H + \rho I)^{-1},$$

up to the equivalent ordering convention for the commutator, and then bound the four resolvents by their smallest shifted eigenvalues. $\square$

For registration,

$$[H, G]_{ij} = (G_0)_{ij} (H_i - H_j),$$

so local coefficient differences control noncommutativity. For the rank-one model and nonzero gradients, set $\hat{q}_i = q_i / \|q_i\|_2$ and $P_i = \hat{q}_i \hat{q}_i^\top$. The sign-invariant orientation difference between two rank-one Hessians is

$$\|P_i - P_j\|_2 = \sqrt{1 - (\hat{q}_i^\top \hat{q}_j)^2} = |\sin(\phi_i - \phi_j)|,$$

where orientation angles are taken modulo π . This projector metric, unlike $|\sin((\phi_i - \phi_j)/2)|$, is invariant under either sign change $q_i \mapsto -q_i$. The archived controlled runs did not store a normalized commutator or projector-distance summary, so no quantitative claim based on such a statistic is made here.

9 Certificate experiments and scaling

9.1 Structured enclosure comparison

The controlled suite contains 18 cases. Here ϱ_{act} is the true spectral radius at the selected parameters, ϱ_{ora} is the radius at numerically oracle-optimized parameters, and U is the stated rigorous enclosure at the selected pair. The enclosure gap is $(U - \varrho_{\text{act}})/\varrho_{\text{act}}$ and the parameter-selection gap is $(\varrho_{\text{act}} - \varrho_{\text{ora}})/\varrho_{\text{ora}}$. A collapse denotes selection at the boundary of the admissible set.

Table 1: Controlled structured-envelope results; medians over 18 cases.

Method	Enclosure gap	Parameter gap	Collapses
Block resolvent	0.166	0.926	6
Block Gershgorin	0.644	0.560	0
Commutator rectangle	0.602	0.431	0
Comparison rectangle	0.595	0.386	0

Every enclosure contained the exact radius. Minimizing the additive bound produced over-conservative relaxation and collapsed in six of 18 cases. The Perron rectangle retained sign information and had the smallest parameter gap among the signed constructions, but was still too loose for online tuning.

9.2 Controlled variable-coefficient predictor tests

The ten 3×3 cases comprise periodic magnitude variation, smooth orientation variation, one sharp orientation interface, and Gaussian-smoothed image gradients; nine are smooth. The four-corner predictor attained median parameter gaps 0.057 overall and 0.027 on the smooth subset, with no collapse; the abrupt-interface gap was 0.193. At the selected parameters, the median absolute relative discrepancy between the four-corner envelope and the true radius was 1.4×10^{-15} , with maximum 2.8×10^{-3} . The Perron comparison certificate was below one in all ten cases. Its median enclosure gap was 1.29, establishing validity while illustrating conservatism. A normalized commutator-magnitude analysis remains TODO because that diagnostic was not stored by the archived runs.

9.3 Truncation scaling

The truncated-kernel certificate showed no violation in 20 tests spanning 8^2 through 64^2 grids and five truncation radii. Actual matrix-free spectral radii were approximately 0.269, while certificate values ranged from 0.638 to 0.681, corresponding to relative gaps 1.38–1.54. The most expensive 64^2 retained-radius case took 81.4 seconds on CPU. A relaxed 128^2 Arnoldi certificate run did not complete within its allocated diagnostic window. The sparse representation scales in principle, but the current eigensolver and bounds are not practical registration-time certificates.

9.4 Additional algebraic verification

Ten thousand randomized curvature configurations were used to compare explicit pixelwise curvature enumeration with the four-corner envelope. No discrepancy was observed. Constant-coefficient periodic systems matched the full reduced spectral radius to numerical precision. These tests validate the implementation of the separable theorem; they do not turn the variable-coefficient predictor into an exact global spectral method.

10 Implementation and baseline details

10.1 Solver

All methods use the same three-level image pyramid, initialization, linearization, stopping tolerances, interpolation, line search, and discrete-Jacobian safeguards. Pixelwise data solves use the Sherman–Morrison formula; regularizer solves use FFT diagonalization. The line search requires objective

decrease and a discrete-Jacobian floor of 0.05. When pyramid upsampling violates this floor, only the nontranslational displacement is halved until the check passes. This numerical safeguard does not constitute a global diffeomorphism theorem.

10.2 Baselines

The external fixed pair $(\rho, \alpha) = (0.1, 1)$ was selected using the controlled public-image pilot, not FIRE or CIMA, and was then frozen. Residual balancing starts at $\rho = 1$, updates the penalty from primal and dual residual imbalance, and rescales the dual variable. The BB comparison is a safeguarded Barzilai–Borwein penalty proxy with clipping and periodic updates. Fixed oADMM uses $(1, 1.8)$. Exact update intervals and safeguards are implemented in `src/registration2d.py`.

10.3 Patch and symbol construction

Optional certificate studies use fixed, multiresolution, or adaptively split patches. A representative $\bar{q}_r$ may be a center, mean, or robust spatial median, and

$$H_r = \nu I_d + \mu \bar{q}_r \bar{q}_r^\top.$$

For a direct block surrogate,

$$\delta_H = \max_i \left\| \mu q_i q_i^\top + \nu I_d - H_{r(i)} \right\|_2.$$

Periodic or cosine closures provide analytic regularizer endpoints. For an order- p Sobolev regularizer,

$$g(\xi) = \left(\gamma + \frac{4\beta}{(\Delta x)^2} \sum_{\ell=1}^d \sin^2(\xi_\ell/2) \right)^p,$$

so

$$g_- = \gamma^p, \quad g_+ = \left(\gamma + \frac{4\beta d}{(\Delta x)^2} \right)^p.$$

11 Additional benchmark information

11.1 Controlled real-content pilot

A public-domain retinal image and an unrestricted immunohistochemistry image were each subjected to five smooth known deformations. Cases 1–2 selected the single external fixed pair; held-out cases 3–5 were not used for tuning and gave six image–deformation tests. Across five order-alternated repeats (30 paired runs), one-shot tuning reduced median inner iterations by 44.4% and median total time by 41.7% (bootstrap 95% interval $[40.9, 42.6]\%$). Selection consumed 1.58% of proposed runtime, median relative MSE change was -0.0068% , and no tested case had a nonpositive Jacobian.

Table 2: Controlled real-content pilot against the predeclared acceptance gates; medians over paired runs.

Metric	Observed	Gate
Inner-iteration reduction	44.4%	$\geq 15\%$
Total-time reduction	41.7%	$\geq 10\%$
Relative MSE change	-0.0068%	$\leq 1\%$
Selection overhead	1.58%	$< 2\%$
Nonpositive Jacobians	0/6	0

11.2 CIMA preprocessing and initialization

The public lung-lesion-3 sample comprises five differently stained sections and all ten unordered pairs. Images supplied at 5% scale were converted to a common structural channel by grayscale conversion, contrast-limited adaptive histogram equalization (CLAHE), Sobel-magnitude extraction, Gaussian smoothing, and fixed percentile normalization, and were then resized to 256^2 . Manual landmarks supplied at 50% scale were rescaled geometrically and used only for evaluation. Each method used the same image-only robust translation initialization selected from phase-correlation, centroid, and clustered SIFT proposals on a 128^2 image.

11.3 FIRE parameter summary

The audited FIRE result contains 134 pairs and five methods per pair. Percentage reductions are computed pairwise before aggregation. For one-shot four-corner tuning, median $\rho_{4C} = 0.1008$ (IQR 0.0397, range 0.0688–0.1402) and median $\alpha_{4C} = 1.5699$ (IQR 0.1287, range 1.4483–1.6883). The median h_+ is 0.2249.

11.4 One-shot reuse

On a balanced 18-pair FIRE subset, per-level retuning reduced median inner iterations from 243 to 222 but increased median tuning time from 0.030 to 0.110 seconds and median total time from 2.651 to 3.895 seconds. Accordingly, all headline experiments use one-shot reuse. Retuning at every outer Gauss–Newton iteration was not tested separately.

12 Reproducibility and availability

The audited runs began at commit `3dc78db3013b9ab2eed80f00b04d4455bc809a55` on 6 August 2026 and used seed 20260806 and `configs/real2d.v1.yaml`. The environment used Python 3.10.9, NumPy 2.2.6, SciPy 1.15.3, pandas 2.3.3, Matplotlib 3.10.9, and scikit-image 0.25.2 on two Intel Xeon Gold 5218 CPUs; the experiments were CPU-only. Timed registrations used one BLAS/OpenMP thread. Exact commands and the complete validation sequence are recorded in `results/real2d.v1/reproducibility_log.md`. Raw pair-level outputs, result schemas, and regeneration scripts are stored alongside that log. FIRE preprocessing uses a foreground-percentile-normalized green channel; CIMA uses the structural-channel pipeline above. Supplied landmarks are used only for evaluation.

FIRE and CIMA are public datasets subject to their original licenses. Raw FIRE images are not redistributed. Preparation scripts, integrity checks, all nonrestricted pair-level measurements, and the certificate implementations (`src/structured_envelopes.py` and the corresponding `experiments/` scripts) are available at <https://github.com/xhxuciedu/admm-registration>.